Ab initio constraints on the melting of silica at high pressures up to 500 GPa Supplemental Information

Ming Geng (耿明) [1] and Chris E. Mohn[1,2,*]
[1] *Centre for Earth Evolution and Dynamics (CEED) and Centre for Planetary Habitability (PHAB), University of Oslo, N-0315 Oslo, Norway and*
[2] *Department of Chemistry and Center for Materials Science and Nanotechnology, University of Oslo, Oslo 0371, Norway*
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TABLE S1: Simulation temperature, average pressure, densities and waiting time, $\tau_{\text{equilibrium}}$, before freezing or melting in two-phase *ab initio* NVT MD runs. The first column "solid phase" refers to the crystal structure in the solid portion of the simulation box before the simulation is launched. Simulations with frozen-in defects in the solid state after the run has terminated is labeled as "defects" (i.e. a single Si is located at an interstitial site) or more "extended defects" (where several atoms have exchanged positions compared to the ideal crystal structure). Typical error-bars in the equilibrium pressures are between 2 and 5 GPa.

solid phase	T (K)	P (GPa)	ρ (g/cm^3)	$\tau_{\text{equilibrium}}$ (ps)	final phase
stishovite	5750	78	4.49	6.0	L
stishovite	5750	69	4.37	5.2	L
stishovite	5750	59	4.24	3.0	L
stishovite	5500	60	4.37	10.4	L
stishovite	5500	55	4.24	3.0	L
stishovite	5500	48	4.12	3.0	L
stishovite	5500	43	4.00	2.0	L
stishovite	5250	55	4.24	9.5	L
stishovite	5250	41	4.00	3.6	L
stishovite	5250	37	3.89	1.9	L
stishovite	5000	45	4.12	5.0	L
stishovite	5500	100	4.93	6.0	S (stishovite w defects)
stishovite	5500	85	4.77	8.8	S (stishovite w extended defects)
stishovite	5500	70	4.64	8.4	S (stishovite)
stishovite	5500	55	4.50	19.4	S (stishovite)
stishovite	5250	100	4.93	8.2	S (stishovite)
stishovite	5250	86	4.77	4.5	S (stishovite)
stishovite	5250	62	4.50	6.6	S (stishovite)
stishovite	5250	58	4.37	5.2	S (stishovite)
β-stishovite	6000	90	4.60	3.0	L
β-stishovite	6000	100	4.74	8.0	L
β-stishovite	6250	100	4.74	2.4	L
β-stishovite	6250	111	4.88	7.2	L
β-stishovite	6250	131	5.03	18.0	L
β-stishovite	6500	146	5.19	5.4	L
β-stishovite	5750	81	4.74	13.8	S (β-stishovite)
β-stishovite	5750	75	4.60	7.8	S (β-stishovite
β-stishovite	6250	141	5.12	8.0	S (β-stishovite w. defect cluster)
β-stishovite	6250	156	5.35	14.0	S (β-stishovite)
β-stishovite	6250	182	5.52	16.4	S (β-stishovite)
β-stishovite	6000	180	5.52	4.2	S (β-stishovite) above stab. field
β-stishovite	6000	158	5.35	8.0	S (β-stishovite)
β-stishovite	6000	136	5.19	4.0	S (β-stishovite)
β-stishovite	6000	117	5.03	5.6	S (β-stishovite w. extended defects)
β-stishovite	6000	102	4.88	5.1	S (β-stishovite)
seifertite	6500	214	5.77	22.0	S (seifertite)
seifertite	6500	141	5.28	11.4	L
seifertite	6750	161	5.28	8.4	L

* chrism@kjemi.uio.no

seifertite	6750	179	5.43	8.3	L
seifertite	7000	180	5.43	6.1	L
seifertite	7000	164	5.28	3.0	L
pyrite	6500	152	5.78	12.6	S (pyrite)
pyrite	6500	178	5.62	7.4	S (pyrite)
pyrite	6750	173	5.32	14.6	S (pyrite)
pyrite	6750	194	5.47	2.0	S (pyrite)
pyrite	6750	218	5.32	2.0	S (pyrite)
pyrite	6750	245	5.17	2.6	S (pyrite)
pyrite	7000	248	5.17	2.8	S (pyrite)
pyrite	7000	224	5.32	4.4	S (pyrite)
pyrite	7250	224	5.94	8.3	S (pyrite)
pyrite	7250	251	5.17	3.0	S (pyrite)
pyrite	7250	289	6.28	4.0	S (pyrite)
pyrite	7500	282	6.28	3.2	S (pyrite)
pyrite seed 2	7500	284	6.28	5.2	S (pyrite)
pyrite	7500	314	6.47	2.3	S (pyrite)
pyrite	7750	319	4.88	4.5	S (pyrite)
pyrite	8000	330	6.47	11.9	S (pyrite)
pyrite	8000	357	6.66	11.1	S (pyrite)
pyrite	8000	399	6.85	4.2	S (pyrite)
pyrite	8000	441	7.06	13.9	S (pyrite)
pyrite	8250	400	6.85	2.6	S (pyrite)
pyrite seed 2	8250	399	6.85	2.8	S (pyrite)
pyrite	8250	443	7.06	14.4	S (pyrite)
pyrite seed 2	8250	443	7.06	5.0	S (pyrite)
pyrite	8500	546	4.21	9.6	S (pyrite)
pyrite	8500	494	4.34	32.5	S (pyrite)
pyrite	7000	206	5.62	5.0	L
pyrite	7250	250	5.32	14.2	L
pyrite	7250	228	5.77	8.8	L
pyrite	7250	205	5.62	2.7	L
pyrite	7500	209	5.62	1.8	L
pyrite	7500	249	5.94	4.4	L
pyrite	7500	229	5.77	1.5	L
pyrite	7500	280	6.11	8.8	L
pyrite	7750	311	5.02	5.0	L
pyrite	7750	283	5.17	2.8	L
pyrite	7750	257	5.32	2.4	L
pyrite	8000	306	6.28	2.7	L
pyrite	8000	286	5.17	2.0	L
pyrite	8250	373	6.65	8.0	L
pyrite seed 2	8250	376	6.65	6.0	L
pyrite	8250	348	4.88	3.0	L
pyrite seed 2	8250	347	4.88	2.9	L
pyrite	8500	385	4.74	3.0	L
pyrite	8500	422	4.61	2.6	L
pyrite	8500	463	4.47	4.6	L